Logic Bubble — Project Management

1. Team

Name	Role
Xuehua Hu	Front-End/Full-Stack Developer
Jing Lyu	Back-End/Full-Stack Developer
Samia Ouagague	UI/UX Designer
Shiran Qiao	UI/UX Designer

Xuehua Hu is responsible for frontend development, overall system architecture design, version management, and coordinating integration between frontend and backend.

Jing Lyu handles backend development using Flask and MySQL, including API routes, database schema, and server-side logic, with contributions to the frontend as well.

Samia Ouagague contributes to UI/UX design — wireframe development, page layout, component visual specifications, and visual consistency across the product.

Shiran Qiao leads the visual design work — branding direction, CSS styling, page layout, and defining the component visual language across the app.

2. Task Plan & Timeline

Date	Task
Feb 25 – Mar 11	Concept discussion and research: brainstormed project directions across multiple sessions, conducted competitor analysis, and confirmed the Logic Bubble concept with defined target users
Mar 12 – Mar 16	Wireframes (editor page): designed the core canvas layout, node anatomy, and interaction patterns
Mar 17 – Mar 21	Wireframes (other pages) + visual design: completed home, dashboard, login/register wireframes; finalised colour palette, typography, and component visual language
Mar 22 – Mar 27	Initial demo + frontend: built the basic canvas (bubble create, drag, delete), set up Flask routing and auth pages
Mar 28 – Apr 5	Backend and frontend development: full auth flow, project save/load with MySQL, SVG curved connections, zoom & pan, snapping, shortcut system, mode switching, UI topbar overhaul, admin panel
Apr 5 – Apr 8	Polish and bug fixes: focus ring fix, straight/curve toggle sync, gear menu fix, UI improvements

Date	Task
Apr 8 – Apr 15	Final fixes and documentation: modifier-key shortcuts, font size controls, settings panel, final testing, submission materials

3. Meeting Notes

Feb 25 — First Brainstorm Kicked off the project. Threw around a bunch of ideas — portfolio site, quiz app, task manager, concept mapper. Nothing decided yet, but everyone left with something to research.

Mar 1 — Second Brainstorm Shared findings from independent research. After several rounds of discussion and considering the project timeline, the team decided to build a tool-based logic diagram product.

Mar 11 — Concept Confirmed After a few rounds of iteration, we confirmed the concept: Logic Bubble, a node-and-connection editor for mapping logical relationships. Core features agreed: create/edit/delete bubbles, draw connections, save/load projects, user accounts. Target users: students, writers, researchers who need quick visual thinking tools.

Mar 14 — Wireframes Started Designers shared early wireframe sketches for the editor canvas. Talked through node layout, connection behaviour, and the dark-first visual style. Decided on a minimal topbar with settings tucked into a sidebar.

Mar 21 — Wireframes Complete All pages done — home, login, register, dashboard, editor. Colour palette confirmed (dark background, blue accents, warm white text). Typography locked in. Design system documented and ready for development.

Mar 24 — First Working Build Basic canvas working: bubbles can be created, dragged, and deleted. No connections yet. Undo/redo stack in place.

Mar 27 — Auth Routes Working Flask routes for register, login, and logout are up. Dashboard pulling a project list from MySQL. Sessions working. Found a bug: deleting a project leaves a blank page — noted for later.

Mar 29 — Backend Complete Full auth flow stable. Project save and load working with MySQL. Dashboard shows real data. Moving on to feature development.

Apr 2 — Major Features In Zoom and pan working. SVG curved connections in. Snapping grid in.

Apr 4 — v4.0 Done Help modal, shortcut system, mode switching (select / connect / delete), right-click context menu all complete. Editor is starting to feel like a real product.

Apr 7 — Topbar and Admin Panel Redesigned the editor topbar — much cleaner hierarchy. Admin panel added with user list and announcement board. Toast notifications unified across all pages. Found several bugs during testing — all logged.

Apr 10 — Bug Sprint Worked through the bug list: focus ring fixed (was affecting global button styles), ESC cancel works in all contexts, logo hover menu timing fixed.

Apr 12 — Nearly Done Mode switching updates visually on the spot. Modifier-key shortcuts in. Font size +/- controls added. Settings panel fully unified. Everything retested.

Apr 15 — Final Testing & Wrap-Up End-to-end test over all flows — auth, editor, admin panel. Two known issues at submission: modifier shortcuts can't be modified via settings panel, and long text in bubbles has a proportion calculation bug. Both documented. Submission materials ready.

4. Version Log

v0.3.0 Early prototype stage. Established the basic project structure: unified UI styling, project CRUD on the dashboard, colour picker, and project export. Migrated JavaScript to native ES Modules and added basic tests.

v1 / v2 v1 introduced the core canvas — bubbles can be created, repositioned, and deleted. v2 added the full authentication flow (register, login, logout, session persistence) and connected the editor to MySQL for project save and load.

v4.0 Major feature build. Added SVG curved connections between nodes, zoom and pan, snapping grid, keyboard shortcut system, help modal, and mode switching (select / connect / delete). This is when Logic Bubble started feeling like a real tool.

v4.1.0 UI polish pass. Added right-click context menu, mode indicator in the topbar, admin panel, and documentation/about pages.

v4.1.1 Overhauled the editor topbar layout. Unified toast notification style site-wide. Added admin announcement board. Login now accepts both username and email.

v4.1.2 Bug fix sprint. Fixed line toggle behaviour, focus ring scope issue, and gear dropdown overflow. Added opacity control in the colour picker, font type selector, and the I-key shortcut for toggling the hint panel.

v4.1.3 Separated focus ring settings for select and connect modes. Fixed ESC key in shortcut recording, logo hover timing, and straight/curve toggle sync. Reorganised the settings panel layout.

v4.1.4 Mode switching now visually updates immediately on click. Improved bubble spring animation. Fixed straight/curve toolbar sync and cleaned up accumulated code debt.

v4.1.5 Added modifier-key shortcut support. Added +/- font size controls with long-press behaviour. Fully unified the settings panel UI across all sub-panels.

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5. Future Roadmap

The following features are identified for future development beyond the current submission:

1. **Dot node drag-insert into existing connection line (A -> dot -> B)**

Allow users to drag a new node onto an existing connection line, automatically splitting the line and inserting the node between the two endpoints.

2. **User custom font file import**

Allow users to upload their own font files (TTF, OTF, WOFF) for use within the editor. Fonts stored per-account and appear alongside system defaults.

3. **Light theme**

A full light-mode visual theme with equivalent contrast ratios and component styles to the existing dark theme. Theme preference saved per user account.

4. **Admin message read/unread status with red dot badge**

Admin messages should display a red dot badge on the dashboard notification icon for users who have not yet read the message.

5. **Account deletion**

Allow users to permanently delete their own account and all associated project data from within account settings.

6. **AI-powered diagram generation from uploaded text or PDF**

Users can upload a document or paste text, and an AI model automatically generates a Logic Bubble diagram with nodes and connections based on the content.

7. **PDF import with in-page reading and annotation**

Users can import a PDF into Logic Bubble, view it in a side panel while building a diagram, and convert annotations into bubbles or links.

8. **Integration into third-party reading applications as an embedded feature**

Logic Bubble's editor embedded within third-party reading and research platforms as an inline component, letting users spawn diagrams directly from highlighted text.